CLASSROOM SCIENCE SERIES

Unlocking Life's Code: DNA Extraction from a Strawberry

Discover how to crush, extract, and visualize real DNA strands right from kitchen materials!

1. What Is DNA?

The Recipe Book of Life

DNA (Deoxyribonucleic Acid) acts as a tiny recipe book inside every living cell. It contains all the instructions that help living organisms grow, function, and look like themselves—from a towering banyan tree to a Bengal tiger!

Cell
The fundamental building block of all life.

Nucleus
The control center of the cell where DNA resides.

DNA
The master recipe book containing genetic code.

Gene
Small individual instructions telling cells what to do.

2. Lab Tools & Materials

To extract DNA from a strawberry, gather the following kitchen-safe lab supplies:

- **Fresh Strawberry:** Rich source of DNA (octoploid, having 8 sets of chromosomes).
- **Ziploc Plastic Bag:** For mechanical crushing of fruit cells.
- **Dishwashing Liquid** / **Soap:** Breaks open cell and nuclear membranes.
- **Table Salt:** Helps DNA strands clump together.
- **Water:** Creates the liquid buffer solution.
- **Tea Strainer** / **Filter:** Separates solid pulp from liquid extract.
- **Clear Glass Container:** For observing chemical reactions and layer separation.
- **Ice-Cold Surgical Spirit** / **Rubbing Alcohol:** Precipitates DNA out of the solution.
- **Toothpick** / **Wooden Skewer:** For spooling out the white DNA threads.

Safety First! Lab Guidelines

- **Adult Supervision:** Always ask an adult to assist with the ice-cold surgical spirit/rubbing alcohol step.
- **Proper Handling:** Alcohol is ice-cold, flammable, and strictly NOT for drinking.
- **Cleanliness:** Wash your hands thoroughly after the experiment and keep your workspace dry.

3. Step-by-Step DNA Extraction Procedure

1. **Mash the Fruit:** Place the strawberry inside a Ziploc bag, seal it tightly, and gently crush it until it becomes a smooth, pulpy mixture.

2. **Mix the Extraction Buffer:** Add a mixture of water, dishwashing soap, and salt to the mashed strawberry bag. Mix gently without creating excess foam.

3. **Strain the Liquid:** Pour the pink mixture through a tea strainer into a clear glass to filter out seeds and large pulp pieces.

4. **Add Ice-Cold Alcohol:** Slowly tilt the glass and pour ice-cold rubbing alcohol down the inner wall to form a distinct top layer over the pink juice.

5. **Observe & Extract:** Watch cloudy white, stringy threads of DNA appear at the boundary between the fruit juice and alcohol. Twirl the strands gently using a toothpick!

4. Scientific Explanation: What's Happening?

DNA is invisible while inside fruit cells because it is dissolved in water. Mashing breaks cell walls. Dish soap dissolves fatty cell membranes, releasing DNA into the solution. Salt neutralizes DNA's electrical charge, allowing strands to stick together. Finally, since DNA is insoluble in cold alcohol, pouring cold alcohol causes it to **precipitate** (clump out) into visible white threads!

5. Compare & Experiment: Trying Other Fruits

You can repeat this exact protocol with other soft fruits like bananas or papayas to compare the yield:

Fruit Tested	Ease of Mashing	Precipitation Layer	Observed DNA Yield
Strawberry	Very Easy	Thick, cloudy white layer	High (Octoploid)
Banana	Easy	Medium cloudy strands	Moderate (Triploid)
Papaya	Moderate	Fine white threads	Moderate

STUDENT WORKSHEET & ASSESSMENT

Practice Questions: DNA Extraction

Name: ____ Date: ___________________________

Section A: Multiple Choice Questions

Q1. What is the primary role of dishwashing soap during the DNA extraction experiment?

(a) To change the color of the fruit juice to pink

(b) To dissolve dirt and kill bacteria on the skin

(c) To break down fatty cell and nuclear membranes to release DNA

(d) To make bubbles that lift DNA to the surface

Q2. Why is ice-cold rubbing alcohol added to the strained fruit mixture?

(a) It melts the DNA strands so they dissolve faster

(b) DNA is insoluble in cold alcohol, causing it to clump together and become visible

(c) It warms up the mixture to speed up the chemical reaction

(d) It turns the strawberry seeds white for easy sorting

Q3. Where is DNA located inside a living plant cell?

(a) Inside the cell wall only

(b) Inside the nucleus

(c) Floating freely in the fruit juice

(d) Inside the seeds only

Q4. Which tool is used to spool and collect the visible white DNA strands from the glass container?

(a) Tea strainer

(b) Ziploc bag

(c) Toothpick or wooden skewer

(d) Measuring cup

Section B: Short Answer & Thought-Provoking Questions

Q5. Fill in the blanks with appropriate scientific terms:

a) The control center of a cell where DNA is stored is called the ______________.

b) Table salt is added to the extraction buffer because it helps DNA strands ______________ together.

c) The white, cloudy threads that appear in alcohol are clumps of ______________.

Q6. Explain why mechanical crushing (mashing) of the strawberry in a Ziploc bag is necessary before adding chemical reagents.

Q7. If you perform this experiment with both strawberries and bananas, why might you observe different amounts of visible DNA?

Q8. What safety precautions must be taken when handling cold surgical spirit or rubbing alcohol during this activity?